An Inverse Reinforcement Learning Algorithm for semi-Markov Decision Processes*

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***Abstract* – In this paper, we study the inverse reinforcement learning (IRL) algorithm for semi-Markov decision processes (SMDPs) with average reward based on the performance sensitivity analysis. By analyzing the structural form of the performance difference formula between any two different policies, we utilize the expert policy to transform the IRL problems of SMDPs into convex optimization problems. Based on two different performance difference formulas, two IRL algorithms are proposed and the algorithms are verified on a grid maze with options. The feasibility and effectiveness of algorithms are illustrated in the simulation results.**

Index Terms – Inverse Reinforcement Learning, SMDP, Reward Function, Performance Sensitivity, Optimization.

I. INTRODUCTION

Reward function [1] plays a significant role in Reinforcement learning (RL) [2] for scientific researches and the practical applications. RL is a process in which the agent keeps interacting with the environment to get feedbacks and improving the behavior later through learning. The main characteristic of reinforcement learning is that the performer will not be informed of the next step, but will evaluate the available action through the reward function which is a scalar to measure the extent of the agent's behavior during the process. Generally, the reward functions in RL are almost designated by experience, it's hard to guarantee the optimality of the final behavior of the agent.

Inverse Reinforcement Learning (IRL) [3] is the opposite process of RL to obtain the reward function according to the expert policy or demonstration. By studying the relevant given resources, the reward function is constructed by corresponding optimization algorithm, which satisfies the condition that the expert policy is optimal among all possible policies. It achieves the automatic construction of the reward function, and thus avoids the subjectivity of man-made judgment.

Researchers have done extensive researches on IRL over the last two decades and have done a lot of experiments and applications. Ng and Russell [3] firstly proposed the application of IRL in Markov decision process, which gave the basis of reward function learning. Abbeel and Ng [4] proposed the method of using the linear combination of known feature basis functions to represent the reward function for IRL in the MDP environment model. Ratliff [5] proposed an IRL method based on maximum marginal planning. Ramachandran and Amir [6] proposed Bayesian IRL algorithm. Syed and Schapire [7] proposed an IRL algorithm based on theory of zero-sum game in game theory, and the optimal result was finally obtained by a min-max problem. At the same time, IRL was successfully applied to the unmanned aerial vehicle navigation [8]. Neu [9] proposed an IRL algorithm based on the natural gradient, which constructed the reward function by updating the weight vector with the natural gradient. Ullman [10] applied the IRL approach to human behavior by observing human activities and studying human behavior, which facilitated the imitation and study of human behavior. IRL was also applied to music creation to produce the similar style of music in [11]. Sriraam introduced a multi-robot inverse reinforcement study, which was the first IRL model based on the average criterion [12], and it pointed out the great research significance of IRL with average reward criterion. Other IRL algorithms have been proposed based on different theories, like maximum entropy [13] and relative entropy [14]. IRL was also applied to various fields such as the optimization of shipboard deck scheduling, active monocular SLAM and AI safety.

IRL has also been widely used in control theory, such as inverse optimization control problems [15], like linear programming IRL in [3]. IRL is basically a kind of optimization problem. From the perspective of the special structure for the performance difference formula, the performance sensitivity-based IRL method in [22] proposed a new analysis framework. However, these IRL algorithms above are almost applied in the environment of MDPs [16], [17], [18]. When it comes to SMDPs [19], [20], [21] which contain obvious time factors, these algorithms are not applicable, and issues for inverse reinforcement learning in SMDPs have not be researched yet.

In this paper, we extend the application of the performance sensitivity-based IRL algorithm [22] with average criterion from MDPs to SMDPs. By analyzing the structural form of the performance difference formula of SMDPs, we obtains the reward function through solving a similar convex optimization problem [23] to that of the MDPs. Finally, simulations and experiments in

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a grid maze with options are implemented to verify the effectiveness of the algorithm.

The paper is organized as follows. Section II describes the basic model of SMDPs. In Section III, two kinds of performance difference formulas for SMDPs are introduced. The analysis of IRL from the perspective of performance sensitivity is presented in Section IV. Then, simulation algorithms in a grid maze with options are implemented in Section V. Finally, Section VI analyzes the method and concludes this paper.

II. SEMI-MARKOV DECISION PROCESSES

SMDP are a generalization of MDP as shown in Fig.1. For any moment, the state of the system belongs to a finite state set $S=\{1,2,...,M\}$.The corresponding finite action space is denoted by A. The mapping between all states and actions forms a control policy μ for the system. The set of stationary Markov policies are denoted as Π_s. Le $\tau_0,\tau_1,...,\tau_n,...$ be the decision making epochs with $\tau_0=0$, $n = 0, 1, 2, \ldots$. X_n denotes the state at time τ_n, and we refer to $\{X_1,X_2,...,X_n,...\}$ as the Embedded Markov chain in the SMDP.

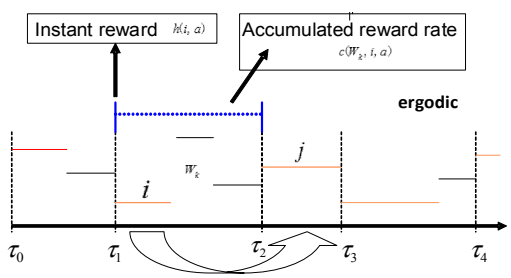

Fig.1 Semi-Markov decision process

At the decision making epoch τ_n, when the system is in state $X_n=i\in S$, an action $A_n=a$ is selected from the admissible action set $A(i)\subseteq A$ according to the current policy. It takes a sojourn time before the next decision-making time τ_{n+1} occurs. The probability that the system moves to state j from state i as a consequence of taking action a within t time units is represented by the following semi-Markov kernel:

$$p(j,t\mid i,a)=P\left(X_{n+1}=j,\tau_{n+1}-\tau_n\le t\mid X_n=i,A_n=a\right) \tag{1}$$

Let $p(j|i,a)$ denote the probability that the embedded Markov chain occupies state j at the subsequent decision epochs when action a is chosen in state i at the current decision epoch, and we have

$$p(j\mid i,a)=p(j,\infty\mid i,a). \tag{2}$$

$F(t\mid i,a)$ denotes the distribution function of sojourn time when state $X_n=i$ and $A_n=a$, then we have

$$F(t\mid i,a)=\sum_{j\in S}p(j,t\mid i,a). \tag{3}$$

The decision making epochs are supposed to be finite in finite time. Between any two decision-making epochs, there is a natural process W_k that the state of the system may vary, where $\tau_n\le k\le\tau_{n+1}$. At each decision time epoch τ_n, if state is i and action is a, the system returns an instantaneous reward $h(i,a)$ and then generates the cumulative reward at rate of $c(W_k,i,a)$ until the next decision. Between any two sequent decision epochs τ_n and τ_{n+1}, the expected total reward and the length of sojourn time are respectively denoted by $r(i,a)$ and $\tau(i,a)$, and we have

$$r(i,a)=h(i,a)+E_i^a\left\{\int_{\tau_n}^{\tau_{n+1}}c\left(W_k,i,a\right)dt\right\}, \tag{4}$$

$$\tau(i,a)=E_i^a\left\{\tau_{n+1}-\tau_n\right\}=\int_0^\infty tF\left(dt\mid i,a\right). \tag{5}$$

where E_i^a denotes the expectation. At any decision epoch when the agent is in state $i\in S$, an action a is chosen from the current policy $w:S\times A\to[0,1]$ with probability $w(a\mid i)$, and we have

$$\sum\nolimits_{a\in A(i)}w(a\mid i)=1. \tag{6}$$

The expected total reward and the expected length of time between two decision epochs under any policy $w\in\Pi_s$ are

$$r^w(i)=\sum_{a\in A}w(i,a)r(i,a),\tau^w(i)=\sum_{a\in A}w(i,a)\tau(i,a). \tag{7}$$

Thus, the SMDP evolves according to semi-Markov kernel $p(j,t\mid i,w)=\sum\nolimits_{a\in A}w(a\mid i)p(j,t\mid i,a)$ and the embedded Markov chain evolves according to the transition probability matrix $P^w=\left[p(j\mid i,w)\right]$ with

$$p(j\mid i,w)=\sum_{a\in A(i)}w(a\mid i)p(j\mid i,a) \tag{8}$$

When policy w is deterministic, the policy specifies an certain action with probability 1 for any state $i\in S$.We denote the policy under this condition as $\nu:S\to A$.Under deterministic policy ν, $p(j,t\mid i,w)$, $p(j\mid i,w)$, $r^w(i)$ and $\tau^w(i)$ are respectively $p(j,t\mid i,\nu(i))$, $p(j\mid i,\nu(i))$, $r(i,\nu(i))$ and $\tau(i,\nu(i))$. Π_d denotes the set of all deterministic stationary policies, and it's obvious that $\Pi_d\subset\Pi_s$.

The embedded Markov chain $\{X_1,X_2,...\}$ is assumed to be ergodic under an arbitrary policy $w\in\Pi_m$.Let $\omega^w=[\omega^w(1),\omega^w(2),...,\omega^w(M)]^T$ represent the steady-state probability of the embedded Markov chain, and we have

$$\omega^wP^w=\omega^w,\omega^we=1, \tag{9}$$

where e denotes a column vector whose all components are 1. We consider the infinite-horizon average reward defined as

$$\rho^w=\frac{\omega^wr^w}{\omega^w\tau^w}. \tag{10}$$

If policy w^* satisfies $\rho^{w^*} \geq \rho^w$ for all policies $w \in \Pi_m$, then it's called the optimal policy and the corresponding average reward ρ^{w^*} is called the optimal reward.

III. TWO PERFORMANCE DIFFERENCE FORMULAS

We intend to find a reward function ensuring that the given expert policy μ_E is optimal among all policies $\mu \in \Pi_s$. To get the characteristics of the reward function, we analyze the difference formula of the performance criteria function between any two different policies.

An equivalent infinitesimal generator $A^\mu = [A^\mu(j \mid i)]$, $i, j \in S$, was defined in [16] for the SMDP introduced in section Π, and it is expressed as

$$A^\mu = \Gamma^\mu [P^\mu - I], \tag{11}$$

where Γ^μ is a diagonal matrix with diagonal elements $\frac{1}{\tau^\mu(1)}, \frac{1}{\tau^\mu(2)}, \ldots, \frac{1}{\tau^\mu(M)}$, and I is an identity matrix. An average-reward performance difference formula based on the infinitesimal generator under different policies $w, v \in \Pi_s$ was given in [16] as follows,

$$\rho^w - \rho^v = q^w [\Gamma^w r^w - \Gamma^v r^v + (A^w - A^v) g^v], \tag{12}$$

where q^w is the state-steady distribution of A^w, i.e.,

$$q^w A^w = 0, q^w e = 1, \tag{13}$$

and g^v is the performance potential, which satisfies the following Poisson equation:

$$A^v g^v = -\Gamma^v r^v + \rho^v e. \tag{14}$$

From [24], there is another performance difference formula in the following form,

$$\rho^w - \rho^v = \frac{\pi^w}{\pi^w \tau^w} \left\{ \left(r^w - r^v\right) - \left(\tau^w - \tau^v\right) \rho^v + \left(P^w - P^v\right) g^v \right\}. \tag{15}$$

Where performance potential g^v satisfies

$$g^v = r^v - \rho^v \tau^v + P^v g^v. \tag{16}$$

It can be proved that this is a variation of (14). Thus, we have

$$g^v = (I - P^v + \mathrm{e}\omega^v)^{-1} (r^v - \rho^v \tau^v). \tag{17}$$

Let $N^w = \frac{\pi^w}{\pi^w \tau^w}$, then we have

$$\rho^w - \rho^v = N^w \left\{ \left(r^w - r^v\right) - \left(\tau^w - \tau^v\right) \rho^v + \left(P^w - P^v\right) g^v \right\}. \tag{18}$$

Equations (12) and (17) are the two different performance difference formulas.

IV. IRL BASED ON SENSITIVITY OPTIMIZATION

IRL uses the expert policy to rebuild the reward function or directly obtain the reward values. Let μ_E be the given expert policy, we expect that it is the optimal among all policies, that is to say, it satisfies

$$\rho^{\mu_E} \geq \rho^\mu, \quad \mu \in \Pi_s, \tag{19}$$

then we have $\rho^{\mu_E} - \rho^\mu \geq 0$.

There is an ill-posed problem in the process of constructing a reward function by the expert policy. As is shown in the figure 2, there may be many reward function values, including the zero reward function, which satisfy the condition that the expert policy is the optimal policy, and this can't determine the optimal reward function value.

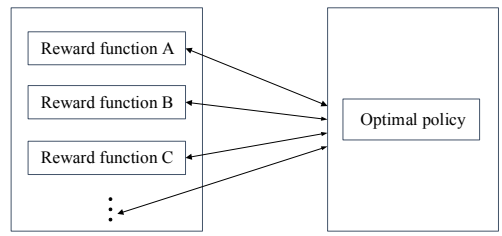

Fig.2 The ill-posed problem of IRL

To avoid the ill-posed problem, the method of maximizing the gap between the optimal policy and sub-optimal policy [3] can be applied to solve the problem. That is to say,

$$\max_r (\rho^{\mu_E} - \max_{\mu \in \Pi_s \setminus \mu_E} \rho^\mu) \tag{20}$$

It can also be written in the following form

$$\max_r (\min_{\mu \in \Pi_s \setminus \mu_E} (\rho^{\mu_E} - \rho^\mu)) \tag{21}$$

From the performance difference formula (12),

$$\rho^\mu - \rho^{\mu_E} = q^\mu [\Gamma^\mu r^\mu - \Gamma^{\mu_E} r^{\mu_E} + (A^\mu - A^{\mu_E}) g^{\mu_E}], \tag{22}$$

That is,

$$\rho^{\mu_E} - \rho^\mu = q^\mu [\Gamma^{\mu_E} r^{\mu_E} - \Gamma^\mu r^\mu + (A^{\mu_E} - A^\mu) g^{\mu_E}]. \tag{23}$$

Since each element of q^μ is nonnegative, and μ_E is the optimal policy if and only if

$$\Gamma^{\mu_E} r^{\mu_E} - \Gamma^\mu r^\mu + (A^{\mu_E} - A^\mu) g^{\mu_E} \geq 0. \tag{24}$$

To make μ_E be the optimal policy, the reward r should satisfy inequality (24). Combining the optimization target and this inequality condition, we obtain the following linear optimization problem for the IRL.

$$\begin{aligned} &\max_r (\min_{\mu \in \Pi_s \setminus \mu_E} ([\Gamma^{\mu_E} r^{\mu_E} - \Gamma^\mu r^\mu + (A^{\mu_E} - A^\mu) g^{\mu_E}]) \\ &s.t. \quad \Gamma^{\mu_E} r^{\mu_E} - \Gamma^\mu r^\mu + (A^{\mu_E} - A^\mu) g^{\mu_E} \geq 0 \\ &\qquad |r(i,a)| \leq R_{\max} \qquad i \in S, a \in A \end{aligned} \tag{25}$$

To ensure that the value function is relatively simple, we add a penalty item $-\lambda_1 \|r\|_1$ to the goal function,

$$\begin{aligned} &\max_r \left\{ \min_{\mu \in \Pi_s \setminus \mu_E} ([\Gamma^{\mu_E} r^{\mu_E} - \Gamma^\mu r^\mu + (A^{\mu_E} - A^\mu) g^{\mu_E}]) - \lambda_1 \|r\|_1 \right\} \\ &s.t. \quad \Gamma^{\mu_E} r^{\mu_E} - \Gamma^\mu r^\mu + (A^{\mu_E} - A^\mu) g^{\mu_E} \geq 0 \\ &\qquad |r(i,a)| \leq R_{\max} \qquad i \in S, a \in A \end{aligned} \tag{26}$$

Based on the convex optimization theory [23], the above optimization problem can be transformed into the following convex optimization problem:

$$\begin{aligned} &\max_r \textstyle\sum_{i=1}^{M}\{t_i-\lambda_1\|r\|_1\} \\ &s.t.\quad \Gamma^{\mu_E}r^{\mu_E}-\Gamma^{\mu}r^{\mu}+(A^{\mu_E}-A^{\mu})g^{\mu_E}\ge t \\ &\qquad |r(i,a)|\le R_{\max}\qquad i\in S\,,a\in A; \end{aligned} \tag{27}$$

where $t=(t_1,\cdots,t_i,\cdots t_M)$.

From the performance difference formula (18), each element of N^w is nonnegative, so the following optimality condition can be obtained. A policy μ_E is the average-reward optimal policy if and only if for any policy $\mu\in\Pi_s$,

$$r^{\mu_E}-\rho^{\mu_E}\tau^{\mu_E}+P^{\mu_E}g^{\mu_E}\ge r^{\mu}-\rho^{\mu_E}\tau^{\mu}+P^{\mu}g^{\mu_E}\,. \tag{28}$$

then the second optimization problem becomes

$$\begin{aligned} &\max_r\left\{\min_{\mu\in\Pi_s\setminus\mu_E}([r^{\mu_E}-r^{\mu}+(P^{\mu_E}-P^{\mu})g^{\mu_E}-(\tau^{\mu_E}-\tau^{\mu})\rho^{\mu_E}])\right\} \\ &s.t.\ r^{\mu_E}-r^{u}+(P^{\mu_E}-P^{\mu})g^{\mu_E}-(\tau^{\mu_E}-\tau^{\mu})\rho^{\mu_E}\ge 0 \\ &\qquad |r(i,a)|\le R_{\max}\qquad i\in S\,,a\in A. \end{aligned} \tag{29}$$

Adding the same penalty item $-\lambda_2\|r\|_1$ to the goal function, (29) is transformed into the following convex optimization problem,

$$\begin{aligned} &\max_r\left\{\textstyle\sum_{i=1}^{M}\left(x_i-\lambda_2\|r\|_1\right)\right\} \\ &s.t.\quad r^{\mu_E}-r^{\mu}+(P^{\mu_E}-P^{\mu})g^{\mu_E}-(\tau^{\mu_E}-\tau^{\mu})\rho^{\mu_E}\ge x\,. \\ &\qquad |r(i,a)|\le R_{\max}\qquad i\in S\,,a\in A \end{aligned} \tag{30}$$

where $x=(x_1,\cdots,x_i,\cdots x_M)$.

The two final optimization problems (27) and (30) can be solved with the Convex Optimization Toolkit CVX. The two algorithms based on different performance formulas for SMDPs are basically the same in the structure, so we here only give the specific steps of the IRL algorithm based on the performance difference formula (12).

IRL algorithm for SMDP

1. Given the expert policy μ_E, set iteration times $T=T_1$.
2. Calculate the corresponding transition probability P^{μ_E}, diagonal matrix Γ^{μ_E} and the equivalent infinitesimal generator A^{μ_E} for expert policy μ_E, $A^{\mu_E}=\Gamma^{\mu_E}[P^{\mu_E}-I]$, randomly choose an initial policy μ_1 and set $k=1$.
3. **FOR** $1\le k\le T_1$ **DO**
4. Calculate the corresponding transition probability, diagonal matrix Γ^{μ_k} and the equivalent infinitesimal generator A^{μ_k} for policy μ_k.
5. Solve the corresponding optimization problem

$$\begin{aligned} &\max_r \textstyle\sum_{i=1}^{M}\{t_i-\lambda_1\|r_k\|_1\} \\ &s.t.\quad \Gamma^{\mu_E}r^{\mu_E}-\Gamma^{\mu_l}r^{\mu_l} \\ &\quad +(A^{\mu_E}-A^{\mu_l})(I-P^{\mu_E}+\mathrm{e}\,\omega^{\mu_E})^{-1}\left[I-\frac{\tau^{\mu_E}\omega^{\mu_E}}{\omega^{\mu_E}\tau^{\mu_E}}\right]r^{\mu_E}\ge t, \\ &\qquad |r_k(i,a)|\le R_{\max},\quad i\in S\,,a\in A,\ l=1,2,\cdots,k \end{aligned}$$

in which we use $g^{\mu_E}=(I-P^{\mu_E}+\mathrm{e}\,\omega^{\mu_E})^{-1}\left[I-\frac{\tau^{\mu_E}\omega^{\mu_E}}{\omega^{\mu_E}\tau^{\mu_E}}\right]r^{\mu_E}$ according to (17).

6. Use policy iterations or value iterations to obtain the optimal policy μ_{k+1} for the SMDP with the constructed reward function r_k.
7. **END FOR**

V. SIMULATION

The feasibility of the above performance sensitivity-based IRL algorithm for SMDPs is verified in the following grid maze with hallway options. As shown in Figure 3, there are four corresponding rooms with hallway A, B, C and D which connects adjacent rooms respectively. For each room, we create 2 options for the agent to move towards different target hallways. The MDP grid environment together with the set of options formally comprises an SMDP.

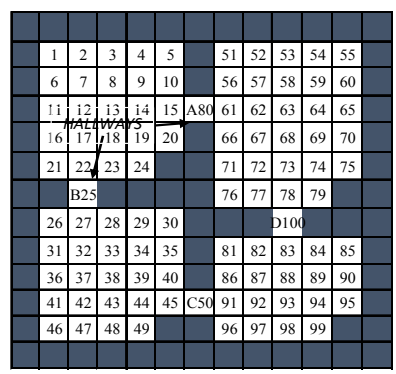

Fig.3 Grid maze with options

The agent can perform up and down about four different actions to move to a different location in each grid cell. Just as the different distances from the grids to the target hallways, the durations of these options are also different. Grids in each room have two options to choose from, and the ending point is D, the situation is simplified since the goal grid is exact one of the terminals for the options. The probability for agent to move to corresponding destination successfully is 0.8.Let H denote the set of 8 different options plus the ninth action to stay still in the goal grid D, which is exactly the whole action space. The policies for the SMDP environment are given in Fig.4. The solid lines indicate the expert choices for grids in each room while the dotted lines denote the other option.

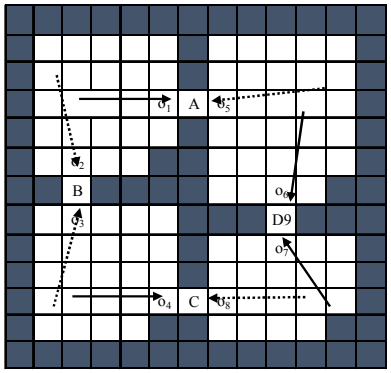

Fig.4 Expert policy among options for the SMDP

For example, the target hallway for option o_1 is A. The inner expert choices for girds in upper left corner room are shown in Fig.5. Then we use the above performance-sensitivity based IRL algorithm with two different optimization problems (27) and (30) to rebuild the reward function.

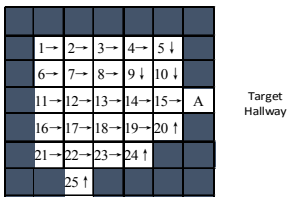

Fig.5 The internal policy for option o_1

The results show the reduced reward function value for each grid cell in Fig.6, Fig.7, Fig.8 and Fig.9 with four different conditions of $\lambda_1 = 0,\ \lambda_1 = 0.1, \lambda_2 = 0, \lambda_2 = 0.15$ respectively. We can see the obvious reward values in the hallway states 25, 50, 80 and 100, respectively.

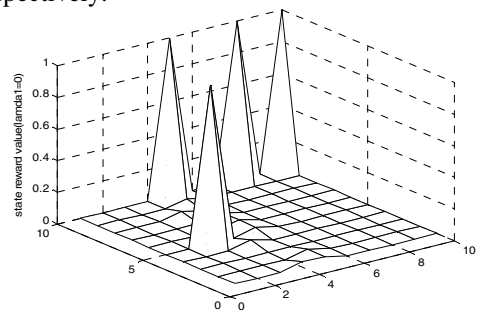

Fig.6 The obtained reward function when $\lambda_1 = 0$

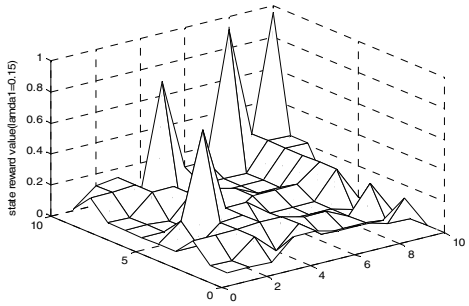

Fig.7 The obtained reward function when $\lambda_1 = 0.1$

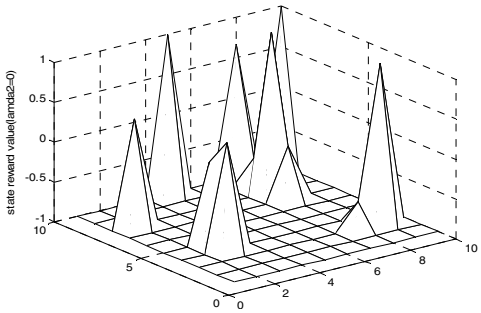

Fig.8 The obtained reward function when $\lambda_2 = 0$

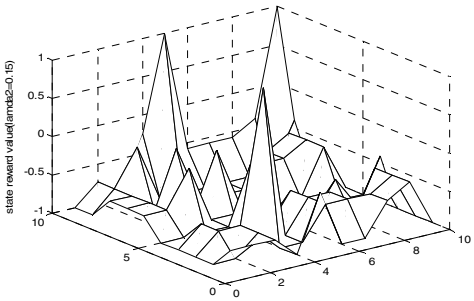

Fig.9 The obtained reward function when $\lambda_2 = 0.15$

Curves for the changes of the number for different policies between the expert policy and the obtained optimal policy and the changes on performance difference for each iteration are shown in Fig.10 and Fig.11.We can see that the algorithm works better under condition of $\lambda_1 = 0.1$ than under $\lambda_1 = 0$ in the convergence process.

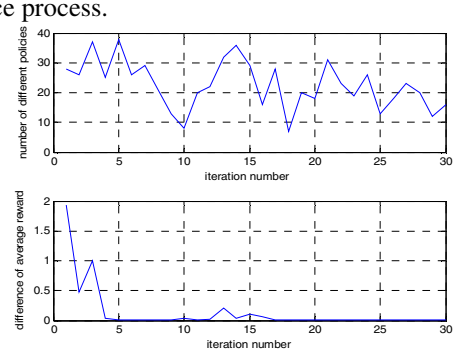

Fig.10 Error analysis with λ_1=0

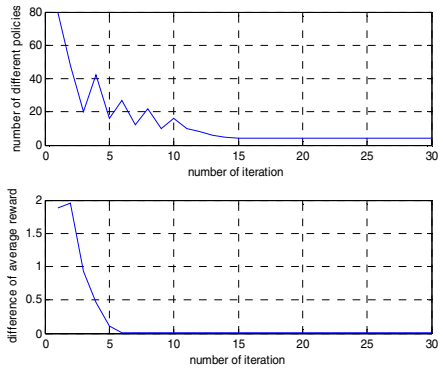

Fig.11 Error analysis with λ_1=0.1

We get the optimal policy according to the final obtained reward function when $\lambda_1 = 0.1$ and $\lambda_2 = 0.15$ .The policies among the action space are as Fig.12 and Fig.13, in which the numbers represent actions that have corresponding subscripts.

	1	1	1	1	1		5	6	6	6	6
	1	1	1	1	1		6	6	6	6	6
	1	1	1	1	1	6	6	6	6	6	6
	1	1	1	1	1		6	6	6	6	6
	2	2	1	1			6	6	6	6	6
		4					6	6	6	6	
	3	3	4	4	4				9		
	3	4	4	4	4		7	7	7	7	7
	4	4	4	4	4		7	7	7	7	7
	4	4	4	4	4	7	7	7	7	7	7
	4	4	4	4			7	7	7	7	

Fig.12 optimal policy when λ_1=0.1

	1	1	1	1	1		5	6	6	6	6
	1	1	1	1	1		5	6	6	6	6
	1	1	1	1	1	6	6	6	6	6	6
	1	1	1	1	1		6	6	6	6	6
	1	2	1	1			6	6	6	6	6
		1					6	6	6	6	
	4	3	4	4	4				9		
	4	4	4	4	4		7	7	7	7	7
	4	4	4	4	4		7	7	7	7	7
	4	4	4	4	4	7	7	7	7	7	7
	4	4	4	4			7	7	7	7	

Fig.13 optimal policy when λ_2=0.15

VI. CONCLUSION

The method presented in this paper extends the performance-sensitivity based IRL theory from MDPs to SMDPs and the feasibility and effectiveness in the SMDP are verified by a grid maze with options. The simulation in this paper is simplified because the goal of the SMDP planning is just one of the options' terminals, and conditions with the goal inside the room will be considered in the future. The reward function values in the algorithm are directly rebuilt by convex optimization theory, it's intuitive and suitable for small state space. For the inverse reinforcement learning problems with large state space, the application of the performance-sensitivity based IRL method needs further improvement, and it may be combined with the existed function approximation.

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